Experiment 5

Sumit Roy
23MS090

PH3105
17/10/2025

1 Title

Study of β-spectroscopy.

2 Aim

Our main objective of this experiment is to study the continuous energy spectrum of beta (β) particles by,

- Measuring the β-spectra of a $\boldsymbol{\beta}^+$ emitter (Na^{22}) and a $\boldsymbol{\beta}^-$ emitter (Sr^{90}).
- Plotting the graph of count rate vs kinetic energy of β-particles.
- Determining the characteristic decay energy for both isotopes from their measured spectra.

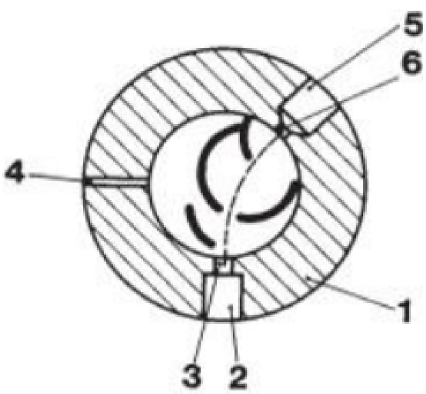
3 Experimental Setup

3.1 Source:

We took Sodium-22 (Na^{22}) a our $\boldsymbol{\beta}^+$ source and Strontium-90 (Sr^{90}) as our $\boldsymbol{\beta}^-$ source.

3.2 β-ray spectrometer:

The main purpose of beta ray spectrometer is to sort the beta particles by their momentum (or kinetic energy). If we use an analogy of the Single channel Analyzer (SCA) used for gamma spectroscopy, here this spectroscope shares almost the same goal in a physical manner. It consists of a cylindrical chamber of non-magnetic material, covered with flat bottom and lid. Three feedthrough are located as the picture, they are for the radiation source, magnetic field probe and exit diaphragm. A predetermined circular arc shaped path consisting of some diaphragms is made inside which leads the sorted β particles from source to exit. A coil wrapped around the spectroscope such that the magnetic field goes inside it along its axis.



Unlike the SCA, here we can't just change the baseline to measure different energies of particles. Instead, we change the magnetic field by changing the current in the coil. As we know, moving charged particles such as β experiences Lorentz force under magnetic field according to the equation:

$$F_{Lorentz} = Bev$$

(Where B is Magnetic field strength, e is charge of particle and v is speed of the particle)
That Lorentz force balences centrifugal force to rotate the β particles' trajectory:

$$Bev = \frac{mv^2}{r}$$

So, the momentum of the particles:

$$p = mv = eBr$$

Thus β particles having the exact momentum to align with the predetermined path through the exit hole, gets to GM counter. As here the e & r is fixed: $p \propto B$ so, we can easily pick up the β particles with a large spectrum of energy just by changing the Magnetic field.

3.3 Geiger Miller (GM) Counter:

The detection of β-particles was carried out using a Geiger–Müller (GM) counter, which converts the ionizing radiation into measurable electrical pulses. It consists...

3.3.1 GM Tube:

The GM tube is a self-extinguishing halogen counting tube for the detection of α, β and γ radiation. It consists of a cylindrical cathode and a central anode wire filled with a low-pressure inert gas. When a radiation is passed through this, it knocks an electron from the outer orbit of that inert gas, this

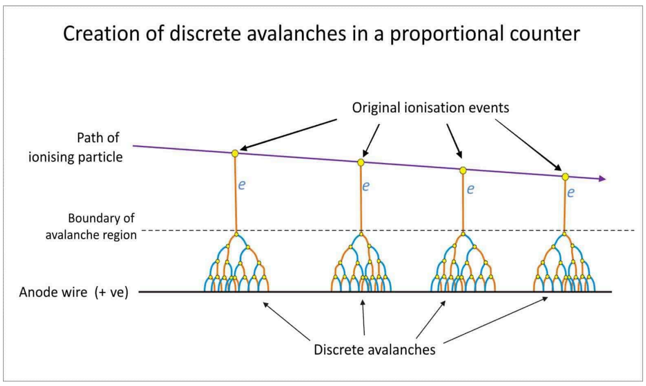


fig: Schematic of Townsend Avalanche

creates an avalanche under the influence of a high electric field. It's called Townsend avalanche which produces a large current pulse.

3.3.2 Counting Circuit :

Each avalanche pulse is counted as a single event by the scaler–timer circuit. This gives the measure of detected beta particles per selected time range.

3.4 Teslameter:

This device measuresMagnetic Field using its thin, flat Hall probe. It operates on Hall effect in which a voltage is developed across a semiconductor probe when a current-carrying conductor is placed in a magnetic field. This Hall voltage is proportional to the magnetic flux density B. Before measurements, the probe output was zero-adjusted with the magnetic field turned off. The calibration of B with magnet current I was first obtained by recording readings for increasing and decreasing currents to trace the hysteresis curve**.**

4 Theory

4.1 Beta (β) Decay:

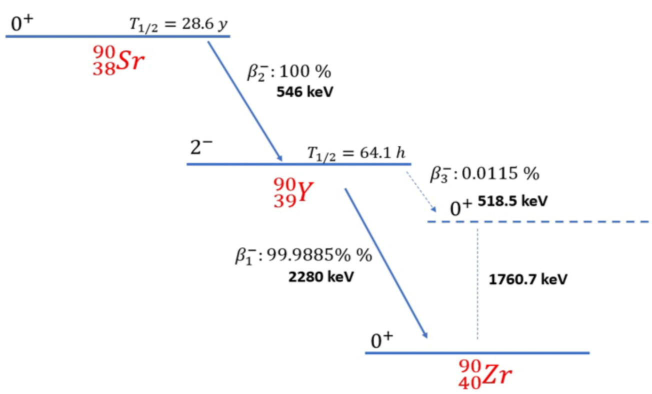
Beta decays also contain decay of neutrino and antineutrino as they carry some energy, The energy left for β particles ranges from zero to the decay energy and we get a spectrum instead of a certain energy value.

4.1.1 β^- Decay:

In β^- decay, a neutron within the nucleus changes into a proton, releasing a beta particle (electron) and an electron antineutrino which is difficult to detect.

$$n \rightarrow p + e^- + \bar{v}_e$$

Decay Scheme of $_{38}Sr^{90}$:



Strontium-90 undergoes pure β^- decay with a maximum β energy of 0.546 MeV, emitting an electron and an antineutrino to form Yttrium-90

$$^{90}\mathrm{Sr} \xrightarrow[T_{1/2}=28.8\ \mathrm{y}]{\beta^-,\ E_{\max}=546\ \mathrm{keV}} {}^{90}\mathrm{Y} + \bar{\nu}_e$$

Yttrium-90 further decays by β^- emission with a maximum energy of 2.28 MeV, producing a stable Zirconium-90 nucleus

$$^{90}\mathrm{Y} \xrightarrow[T_{1/2}=64.1\ \mathrm{h}]{\beta^-,\ E_{\max}=2.28\ \mathrm{MeV}} {}^{90}\mathrm{Zr} + \bar{\nu}_e$$

Both are allowed β^- transitions directly to the ground states of the daughters; there are no significant γ-rays emitted, which is why the spectrum is purely β and continuous.

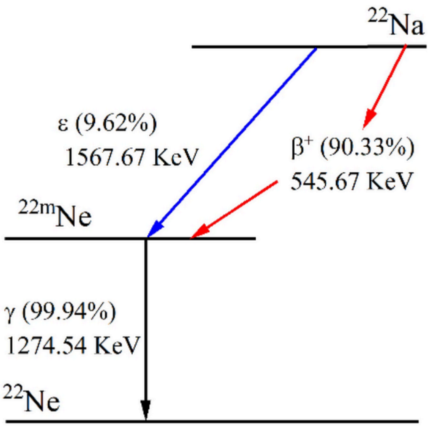
4.1.2 β^+ Decay:

In β^+ decay, a proton transforms into a neutron, releasing a positron (the antimatter version of an electron) and a neutrino.

$$p \rightarrow n + e^- + v_e$$

Decay Scheme of $_{11}Na^{22}$:

A proton in ^{22}Na converts into a neutron, emitting a positron and a neutrino, and leaves the daughter ^{22}Ne nucleus in an excited state.



$$^{22}\mathrm{Na} \xrightarrow[90.3\%]{\beta^+,\ E_{\max}=0.545\ \mathrm{MeV}} {}^{22}\mathrm{Ne}^* + e^+ + \nu_e$$

The nucleus captures an inner-shell electron, turning a proton into a neutron and emitting only a neutrino, again producing excited ^{22}Ne.

$$^{22}\mathrm{Na} \xrightarrow[9.7\%]{EC} {}^{22}\mathrm{Ne}^* + \nu_e$$

The excited ^{22}Ne nucleus de-excites by emitting a 1.275 MeV γ-ray, becoming stable ^{22}Ne.

$$^{22}\mathrm{Ne}^* \rightarrow {}^{22}\mathrm{Ne} + \gamma(1.275\ \mathrm{MeV})$$

4.2 Kinetic Energy of β particle:

Momentum of Beta particle **:** As we have already discussed the working formula of beta spectrometer and got the expression of momentum by equating Lorentz force with centrifugal force

$$p = eBr$$

Relativistic relation of energy-momentum **:** Total energy for relativistic particles (such as β particles) is given by

$$E^2_{Tot} = (pc)^2 + (m_0c^2)^2$$

Where m_0 is the rest mass of the electron (9.109×10^{-31} kg).

So, The kinetic energy of the particle:

$$E_{kin} = E_{Tot} - m_0c^2$$

Substituting for E_{Tot} :

$$E_{kin} = \sqrt{(pc)^2 + (m_0c^2)^2} - m_0c^2$$

Substituting for p :

$$E_{kin} = \sqrt{(eBrc)^2 + (m_0c^2)^2} - m_0c^2$$

This is our final working formula to determine kinetic energy from magnetic field where,

e = $\pm 1.602 \times 10^{-19}$ C (+ for $\boldsymbol{\beta}^+$, - for $\boldsymbol{\beta}^-$)

r = 50mm = 50×10^{-3}m

m_0 = 9.109×10^{-31} kg

c = 3×10^8 m/s

5 Data:

5.1 Sr^{90} decay table:

Coil Voltage (Volts)	Coil Current (Amp)	Mag. Field (mT)	Counts (for 100 seconds)			Kinetic Energy (kev)
			Count 1	Count 2	Average Counts	
0	0	0.3	239	217	228	0.019784994
0.24	0.1	7.3	311	329	320	11.58403075
0.49	0.2	14.4	458	443	450.5	43.71809028
0.75	0.3	22.8	747	741	744	103.7610164
0.98	0.4	31.3	1097	1119	1108	182.743623
1.23	0.5	39.9	1569	1460	1514.5	275.7110843
1.51	0.6	49.6	1857	1913	1885	391.2619915
1.71	0.7	58	2171	1999	2085	497.6044715
1.95	0.8	67.4	2225	2183	2204	621.395858
2.21	0.9	75.9	2257	2337	2297	736.4818395

2.46	1	87	2172	2180	2176	890.0089174
2.96	1.19	101.2	1873	1877	1875	1090.195828
3.48	1.4	117.5	1460	1428	1444	1323.54742
4.25	1.7	140.7	835	849	842	1659.914405
5	2	161.8	507	524	515.5	1968.623005

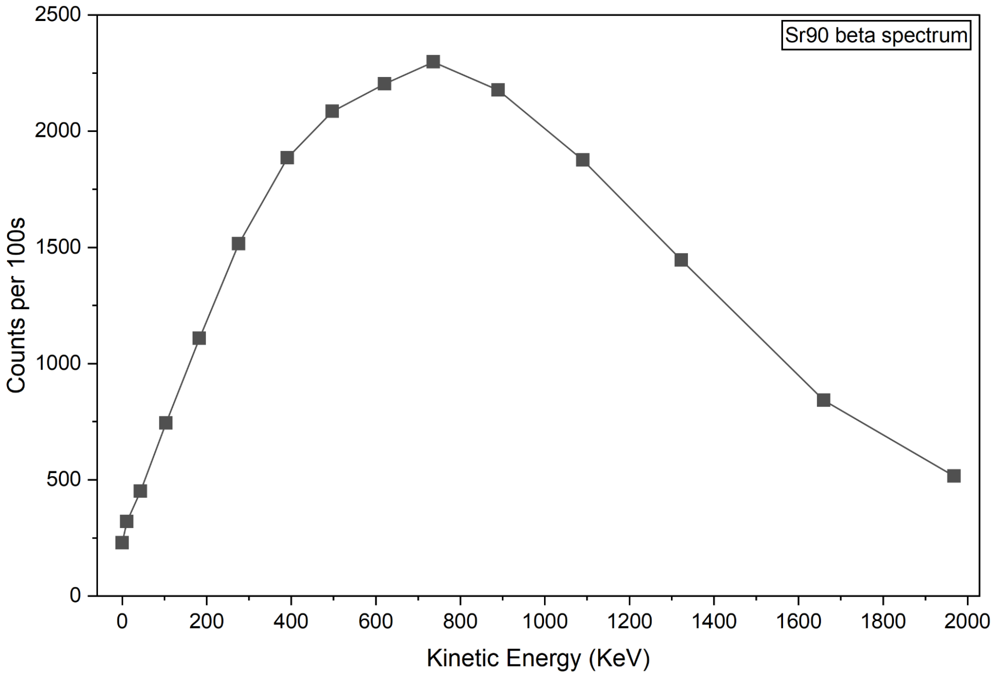
5.2 Na^{22} decay table:

Coil Voltage (Volts)	Coil Current (Amp)	Mag. Field (mT)	Counts (for 100 seconds)			Kinetic Energy (kev)
			Count 1	Count 2	Average Counts	
0	0	-0.5	79	65	72	0.054956429
0.53	0.1	-0.65	96	77	86.5	0.092872926
0.78	0.2	-13	84	85	84.5	35.89374136
1.41	0.3	-21	124	124	124	89.1780868
1.66	0.41	-30.1	171	157	164	170.7038003
1.96	0.5	-39.6	172	147	159.5	272.2963327
2.3	0.6	-49.2	172	137	154.5	386.3249265
2.51	0.7	-61	121	79	100	536.6402456
2.68	0.8	-66.6	107	96	101.5	610.7023844
3	0.9	-76.5	71	74	72.5	744.6961244
3.22	1.01	-87	64	53	58.5	890.0089174
3.02	1.11	-100.4	62	65	63.5	1078.829217
3.29	1.2	-106.8	45	48	46.5	1170.008538
3.8	1.3	-121.2	46	52	49	1376.909649
3.75	1.42	-124.5	46	53	49.5	1424.60439

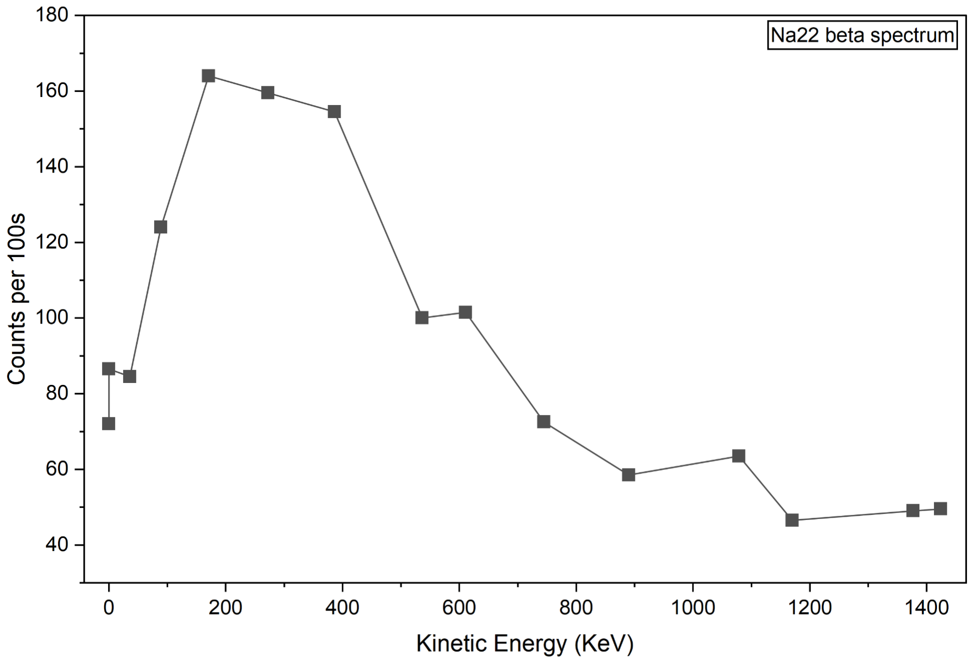
6 Analysis:

6.1 Plots:

6.1.1 Sr^{90} β spectrum plot :



6.1.2 Na^{22} β spectrum plot :



6.2 Results: From our plotted graph, we determined the peaks and found out the most probable energies

6.2.1 Most probable energy(E_p):

Isotope	Type of Decay	Observed Most Probable Energy (E_p)
Sr-90	β^-	≈ 736 keV
Na-22	β^+	≈ 171 keV

6.2.2 Endpoint Energy

Isotope	Type of Decay	Estimated Endpoint Energy (E_{max} ≈ $3E_p$)	Literature E_{max}
Sr-90	β^-	2208 keV	2280 keV
Na-22	β^+	513 keV	545 keV

6.3 Error Analysis: Theoretical Values of the Endpoint Energy didn't exactly match with our estimated values. For Sr^{90} we got 3.15 % error and for Na^{22} we got 5.87% error in Endpoint Energies.

7 Discussion:

The sources had shown a continuous graph of energies with counts, that confirms correct spectrometer behavior. Also we got reasonable results. Still some subtle things like the rough graph of Na-22 and the deviation of E_{max} with literature values indicates slight error prone experiment.

7.1 Sources of error:

- Statistical fluctuations for limited counting time, less number of counts etc.
- Magnetic field calibration error (hysteresis).
- Improper source: Our Na-22 source was a bit error-prone, as the values of counts are fluctuating.
- Background radiation interference.

8 Conclusion:

We have successfully completed the beta spectroscopy using Isotopes of Strontium and Sodium. We obtained the most probable Energies and then derived the end point energies from it. The most probable energies found were consistent with their actual decay characteristics.